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Declaration

I hereby declare that the project work entitled Payroll system submitted to the Bangladesh University of Business and Technology is a record of and original work done by us under the guidance of Nourin Khandaker Mam.

It has not been submitted before for any degree or examination to any other University.

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Chapter One

Introduction

- 1.1 Introduction
- 1.2 Problem Statement
- 1.3 Motivation
- 1.4 Project Objectives
- 1.5 Contribution
- 1.6 Feature
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1.1 Introduction

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Café Management System project is not an internet based. But customers can ordered food very easily. It's significant increase in customers due to new renovations in the local area. There is now easy access to parking and an improved streetscape. This unpredicted change and instant increase in customers has caused problems at the café.

The orders are taken verbally and there is no proper system to manage the orders and stocks. Some of the problems are increased waiting time for the customers, orders taking longer to prepare, incorrect stock levels, inadequate stock control etc. Several complaints have been coming verbally through customers and via social media.

This project developed by C# (C Sharp) programming language and MySQL for stored data.

1.2 Problem Statement

The main aim of this project to save valuable time of customers. Time is very important for us. So, everyone can order food at the same time.

Recent time when people visit a restaurant they have waited for a waiter. But for using this system customers can easily order food and pay their bill. It's totally secure for every customer. So no tension about security. Customers can collect their receipt when they pay their bill. Customers can pay their bill in cash, credit/debit card or other money transfer application. So, this is our facilities of the project. There are no difficulties. Everyone can use our system. Have a nice day.

1.3 Motivation

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The world is moving so fast. The world is now fully depended on technologies. So, we think that to create this project. Everything get easier day by day. We inspired this project form other project. But our project isn't a copy project of them. It's totally not unique but different project. So, we create this project.

1.4 Project Objectives

The project fully technology based. Majority people use technology. So, we create this and give our customer this opportunity.

Primary Objectives

First of all our target is to create everything in easily. Customers can order food very easily. Customers can take their receipt easily. Customer's information are secured in our database. Without our manager/admin no one can see customer's information even not a waiter or chef. As a customer, they can see our product, price and search the product.

Secondary Objectives

We have to learn about C Sharp and Object Oriented Program (OOP). Basically, this program used for create a class and its object. In this project this program developed in this object oriented program. The entire features are created on OOP. C Sharp is programming language. It's not easy language. But not hard language.

1.5 Contribution

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The main aim for the project to us. How to get easier for customers. It contributes to us and the organization in different ways.

- ❖ Firstly it contributes for customers. Customers can order some food very easily. Customers can pay their bill in 3 ways (cash, debit/credit, other money transfer application). They can collect their receipt after paying bill.
- ❖ Secondly it contributes for the organization. The manager of this company they easily access all the employees' information, customers information. Employees can add product, update product, remove product very easily. Waiter and chef can add their information, update their information.
- ❖ Finally this project contributes for us. We can get knowledge about C Sharp, MySQL, OOP. This language we learn in theoretical. But this time we learn this practically. So, this project important for us.

1.6 Conclusion

In this project user can know about our basic information of our project. Section 1.1 introduced about this project. Section 1.2 users can know about our aim, our vision. Section 1.3 is our motivation, inspiration of this project. Section 1.4 is our project objectives and section 1.5 is contribution of this project to us, to customers and others...

Technology

2.1 Introduction

2.2 Database

2.3 C Sharp

2.4 Schema Diagram

2.5 ER Diagram

2.6 Xampp

2.7 Microsoft Visual Studio 2019

2.8 Conclusion

2.1 Introduction

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This section or chapter is especially for an engineer. Because to create this projects this language are used C Sharp and MySQL. This language is used for create a project. But MySQL Server is used for create database. Here we store all the information. Like user information, product information etc. In previous section we told about OOP and C Sharp. Now we will told about this.

2.2 Database

A database is an organized collection of data stored and accessed electronically from a computer system. Where database are more complex they are often developed using formal design and modeling techniques.

Formally, a "database" refers to a set of related data and the way it is organized. Access to this data is usually provided by a "database management system" (DBMS) consisting of an integrated set of computer software that allows users to interact with one or more databases and provides access to all of the data contained in the database (although restrictions may exist that limit access to particular data). The DBMS provides various functions that allow entry, storage and retrieval of large quantities of information and provides ways to manage how that information is organized.

Databases are used for storing, maintaining and accessing any sort of data. They collect information on people, places or things. That information is gathered in one place so that it can be observed and analyzed. Databases can be thought of as an organized collection of information.

Businesses use data stored in databases to make informed business decisions. Some of the ways organizations use databases include the following:

- Improve business processes. Companies collect data about business processes, such sales, and order processing and customer service. They analyze that data to improve these processes, expand their business and grow revenue.
- Keep track of customers. Databases often store information about people, such as customers or users. For example, social media platforms use databases to store user information, such as names, email addresses and user behavior. The data is used to recommend content to users and improve the user experience.

- Secure personal health information. Healthcare providers use databases to securely store personal health data to inform and improve patient care.
- Store personal data. Databases can also be used to store personal information. For example, personal cloud storage is available for individual users to store media, such as photos, in a managed cloud.

Types of databases

There are many types of databases. They may be classified according to content type:

bibliographic, full text, numeric and images. In computing, databases are often classified based on the organizational approach they use.

- ❖ Relational. This tabular approach defines data so it can be reorganized and accessed in many ways. Relational databases are comprised of tables. Data is placed into predefined categories in those tables. Each table has columns with at least one data category, and rows that have a certain data instance for the categories which are defined in the columns. Information in a relational database about a specific customer is organized into rows, columns and tables. These are indexed to make it easier to search using SQL or NoSQL queries

- ❖ Distributed. This database stores records or files in several physical locations. Data processing is also spread out and replicated across different parts of the network.

- ❖ Cloud. These databases are built in a public, private or hybrid cloud for a virtualized environment. Users are charged based on how much storage and bandwidth they use. They also get scalability on demand and high availability. These databases can work with applications deployed as software as a service.

- ❖ Object-oriented. These databases hold data created using object-oriented programming languages. They focus on organizing objects rather than actions and data rather than logic. For instance, an image data record would be a data object, rather than an alphanumeric value.

One way to classify databases involves the type of their contents, for example: bibliographic, document-text, statistical, or multimedia objects. Another way is by their application area, for example: accounting, music compositions, movies, banking, manufacturing, or insurance. A third way is by some technical aspect,

such as the database structure or interface type. This section lists a few of the adjectives used to characterize different kinds of databases.

- ❖ An in-memory database is a database that primarily resides in main memory but is typically backed-up by non-volatile computer data storage. Main memory databases are faster than disk databases, and so are often used where response time is critical, such as in telecommunications network equipment.

- ❖ An active database includes an event-driven architecture which can respond to conditions both inside and outside the database. Possible uses include security monitoring, alerting, statistics gathering and authorization. Many databases provide active database features in the form of database triggers.

- ❖ A cloud database relies on cloud technology. Both the database and most of its DBMS reside remotely, "in the cloud", while its applications are both developed by programmers and later maintained and used by end-users through a web browser and Open APIs.

- ❖ Data warehouses archive data from operational databases and often from external sources such as market research firms.

- ❖ The warehouse becomes the central source of data for use by managers and other end-users who may not have access to operational data. For example, sales data might be aggregated to weekly totals and converted from internal product codes to use UPCs so that they can be compared with ACNielsen data.

- ❖ Some basic and essential components of data warehousing include extracting, analyzing, and mining data, transforming, loading, and managing data so as to make them available for further use.

MySQL

MySQL is an open-source relational database management system (RDBMS).[5][6] Its name is a combination of "My", the name of co-founder Michael Widenius's daughter,[7] and "SQL", the abbreviation for Structured Query

Language. A relational database organizes data into one or more data tables in which data types may be related to each other; these relations help structure the data. SQL is a language programmers use to create, modify and extract data from the relational database, as well as control user access to the database. In addition to relational databases and SQL, an RDBMS like MySQL works with an operating system to implement a relational database in a computer's storage system, manages users, allows for network access and facilitates testing database integrity and creation of backups.

MySQL is free and open-source software under the terms of the GNU General Public License, and is also available under a variety of proprietary licenses. MySQL was owned and sponsored by the Swedish company MySQL AB, which was bought by Sun Microsystems (now Oracle Corporation). In 2010, when Oracle acquired Sun, Widenius forked the open-source MySQL project to create MariaDB.

MySQL has stand-alone clients that allow users to interact directly with a MySQL database using SQL, but more often, MySQL is used with other programs to implement applications that need relational database capability. MySQL is a component of the LAMP web application software stack and others, which is an acronym for Linux, Apache, MySQL, Perl/PHP/Python. MySQL is used by many database-driven web applications, including Drupal, Joomla, phpBB, and WordPress. MySQL is also used by many popular websites, including Facebook, Flickr, MediaWiki, Twitter, and YouTube.

2.3 C Sharp

C# is a general-purpose, multi-paradigm programming language. C# encompasses static typing, strong typing, lexically scoped, imperative, declarative, functional, generic, object oriented (class-based), and component-oriented programming disciplines. C# was designed by Anders Hejlsberg from Microsoft in 2000 and was later approved as an international standard by Ecma (ECMA-334) in 2002 and ISO (ISO/IEC 23270) in 2003. Microsoft introduced C# along with .NET Framework and Visual Studio, both of which were closed-source. At the time,

Microsoft had no open-source products. Four years later, in 2004, a free and open-source project called Mono began, providing a cross-platform compiler and runtime environment for the C# programming language. A decade later, Microsoft released Visual Studio Code (code editor), Roslyn (compiler), and the unified .NET platform (software framework), all of which support C# and are free, open-source, and cross-platform. Mono also joined Microsoft but was not merged into .NET.

As of 2021, the most recent version of the language is C# 10.0, which was released in 2021 in .NET 6.0.

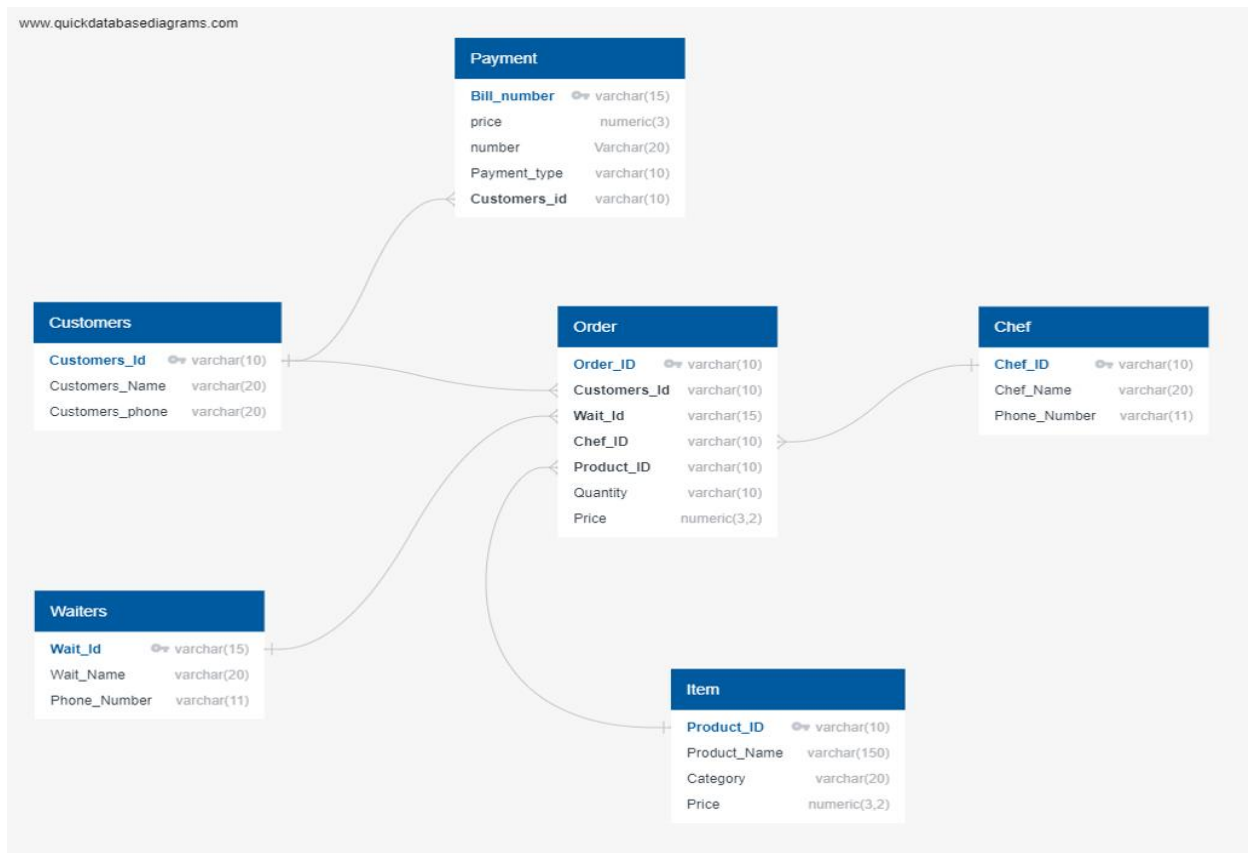
The Ecma standard lists these design goals for C#.

- The language is intended to be a simple, modern, general-purpose, object-oriented programming language.
- The language, and implementations thereof, should provide support for software engineering principles such as strong type checking, array bounds checking, detection of attempts to use uninitialized variables, and automatic garbage collection. Software robustness, durability, and programmer productivity are important.
- The language is intended for use in developing software components suitable for deployment in distributed environments.
- Portability is very important for source code and programmers, especially those already familiar with C and C++.
- Support for internationalization is very important.

2.4 Schema Diagram

A schema diagram is a diagram which contains entities and the attributes that will define that schema. A schema diagram only shows us the database design. It does not show the actual data of the database. Schema can be a single table or it can have more than one table which is related.

Here is our schema diagram :

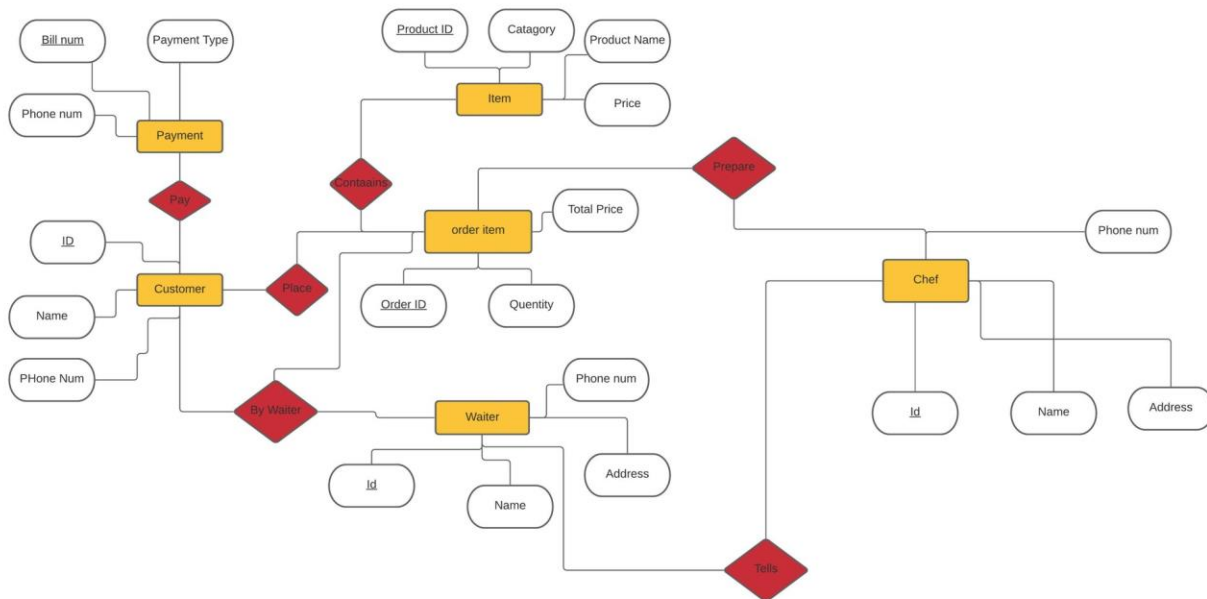


2.5 ER Diagram

An entity–relationship model (or ER model) describes interrelated things of interest in a specific domain of knowledge. A basic ER model is composed of entity types (which classify the things of interest) and specifies relationships that can exist between entities (instances of those entity types).

In software engineering, an ER model is commonly formed to represent things a business needs to remember in order to perform business processes. Consequently, the ER model becomes an abstract data model, which defines a data or information structure which can be implemented in a database, typically a relational database. Entity-relationship modeling was developed for database and design by Peter Chen and published in a 1976 paper,[1] with variants of the idea existing previously.[2] Some ER models show super and subtype entities connected by generalization-specialization relationships,[3] and an ER model can be used also in the specification of domain-specific ontology.

Here is our ER Diagram :



2.6 XAMPP

XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages. Since most actual web server deployments use the same components as XAMPP, it makes transitioning from a local test server to a live server possible.

XAMPP's ease of deployment means a WAMP or LAMP stack can be installed quickly and simply on an operating system by a developer, with the advantage that common add-in applications such as WordPress and Joomla! can also be installed with similar ease using Bitnami.

Letter Meaning of XAMPP:

X = An ideographic letter referring to cross-platform.

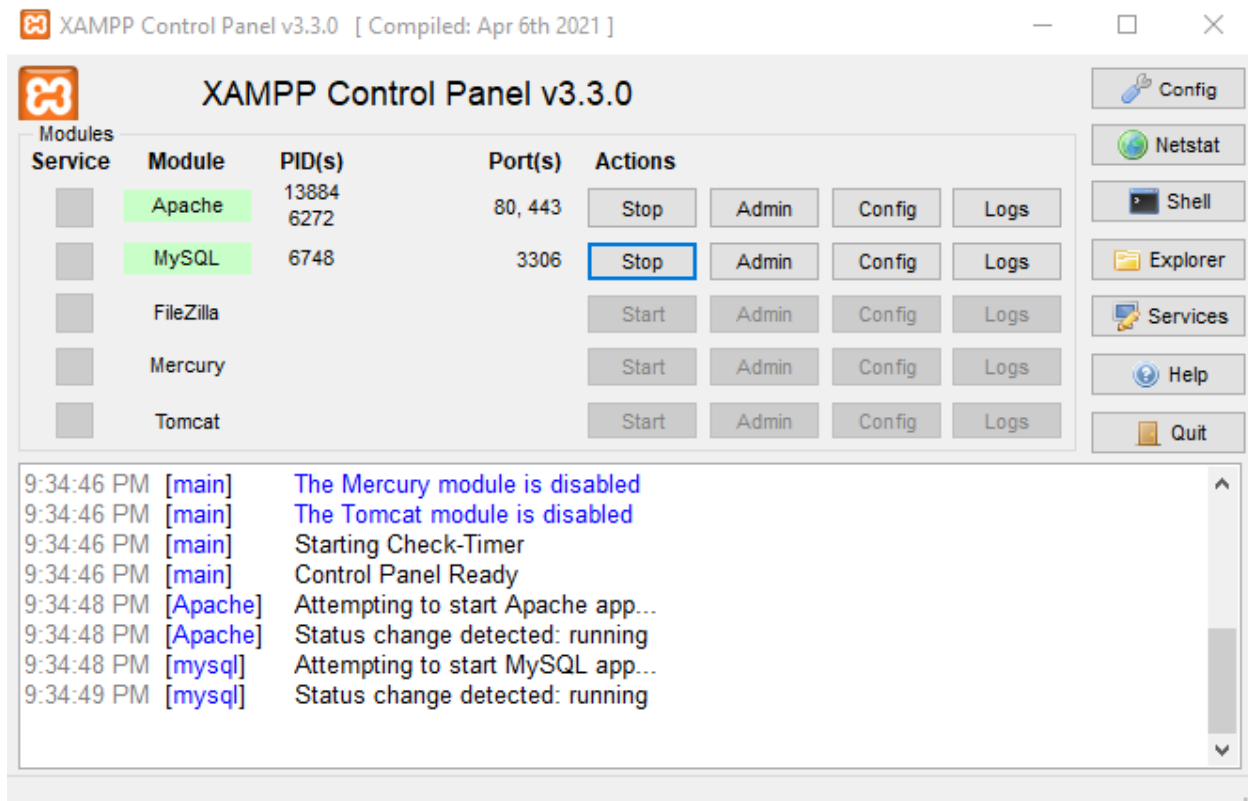
A = Apache, or its expanded form, Apache HTTP Server.

M = MariaDB (formerly MySQL)

P = PHP

P = Perl

MySQL was replaced with MariaDB on 19 October 2015 and beginning with XAMPP versions 5.5.30 and 5.6.14,[7] effectively altering the meaning of the acronym. While both letters P are de facto interchangeable, convention used at the Apache Friends website indicates that the first letter P is short for PHP and the latter letter P is short for Perl.



2.7 Microsoft Visual Studio 2022

Microsoft Visual Studio is an integrated development environment (IDE) from Microsoft. It is used to develop computer programs, as well as websites, web apps, web services and mobile apps. Visual Studio uses Microsoft software development platforms such as Windows API, Windows Forms, Windows Presentation Foundation, Windows Store and Microsoft Silverlight. It can produce both native code and managed code.

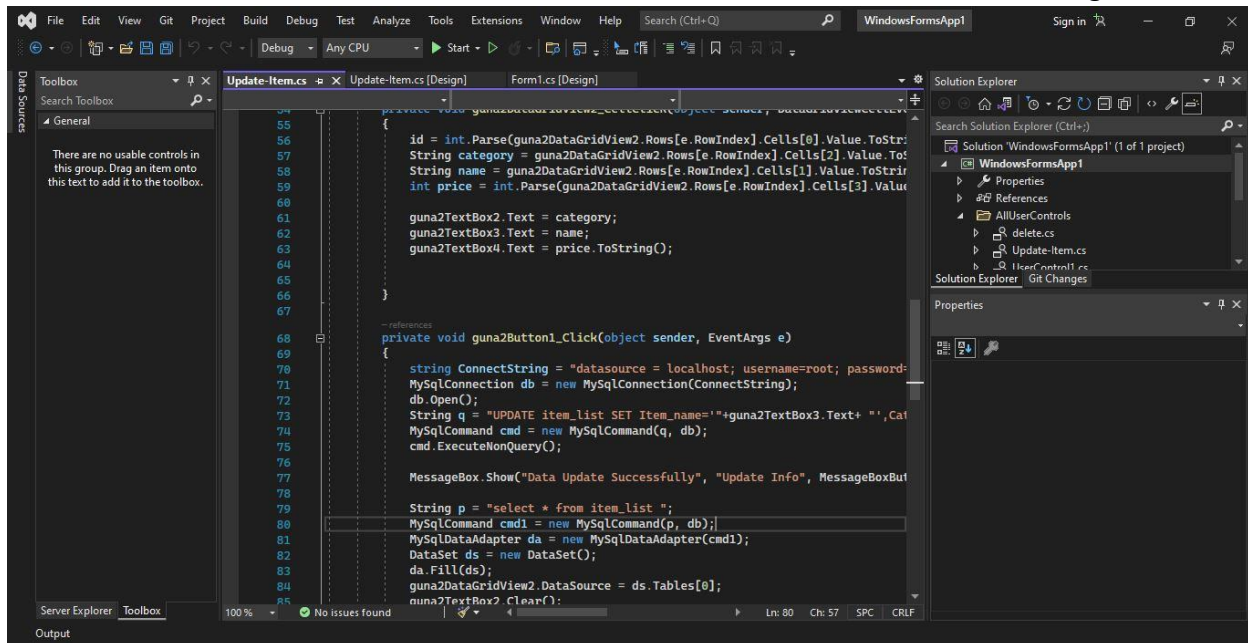
Visual Studio includes a code editor supporting IntelliSense (the code completion component) as well as code refactoring. The integrated debugger works both as a source-level debugger and a machine-level debugger. Other built-in tools include a code profiler, designer for building GUI applications, web designer, class designer, and database schema designer. It accepts plug-ins that expand the functionality at

almost every level—including adding support for source control systems (like Subversion and Git) and adding new toolsets like editors and visual designers for domain-specific languages or toolsets for other aspects of the software development lifecycle (like the Azure DevOps client: Team Explorer).

Visual Studio supports 36 different programming languages and allows the code editor and debugger to support (to varying degrees) nearly any programming language, provided a language-specific service exists. Built-in languages include C, C++, C++/CLI, Visual Basic .NET, C#, F#, JavaScript, TypeScript, XML, XSLT, HTML, and CSS. Support for other languages such as Python, Ruby, Node.js, and M among others is available via plugins. Java (and J#) were supported in the past.

Visual Studio 2022





.Net Framework

The .NET Framework pronounced as "dot net" is a software framework developed by Microsoft that runs primarily on Microsoft Windows. It includes a large class library called Framework Class Library (FCL) and provides language interoperability (each language can use code written in other languages) across several programming languages. Programs written for .NET Framework executes in a software environment (in contrast to a hardware environment) named the Common Language Runtime (CLR). The CLR is an application virtual machine that provides services such as security, memory management, and exception handling. As such, computer code written using .NET Framework is called "managed code". FCL and CLR together constitute the .NET Framework. FCL provides the user interface, data access, database connectivity, cryptography, web application development, numeric algorithms, and network communications. Programmers produce software by combining their source code with .NET Framework and other libraries. The framework is intended to be used by most new applications created for the Windows platform. Microsoft also produces an integrated development environment for .NET software called Visual Studio.

.NET Framework began as proprietary software, although the firm worked to standardize the software stack almost immediately, even before its first release. Despite the standardization efforts, developers, mainly those in the free and open-source software communities, expressed their unease with the selected terms and the prospects of any free and open-source implementation, especially regarding software patents. Since then, Microsoft has changed .NET development to more closely follow a contemporary model of a community-developed software project, including issuing an update to its patent promising to address the concerns. In April 2019, Microsoft released .NET Framework 4.8, the last version of the framework as a proprietary offering. Only monthly security and reliability bug fixes to that version have been released since then. No further changes to that version are planned.

2.8 Conclusion

This chapter is mainly written for an engineer. But we think general people can understand this. Database used for stored data. Schema and ER diagram is our map. XAMPP and Microsoft Visual Studio 2019, .Net Framework is our software. We design this project with that software. That's it.

Chapter Three

System Analysis

3.1 Introduction

3.2 System Analysis

3.3 Interface

3.4 Conclusion

3.1 Introduction

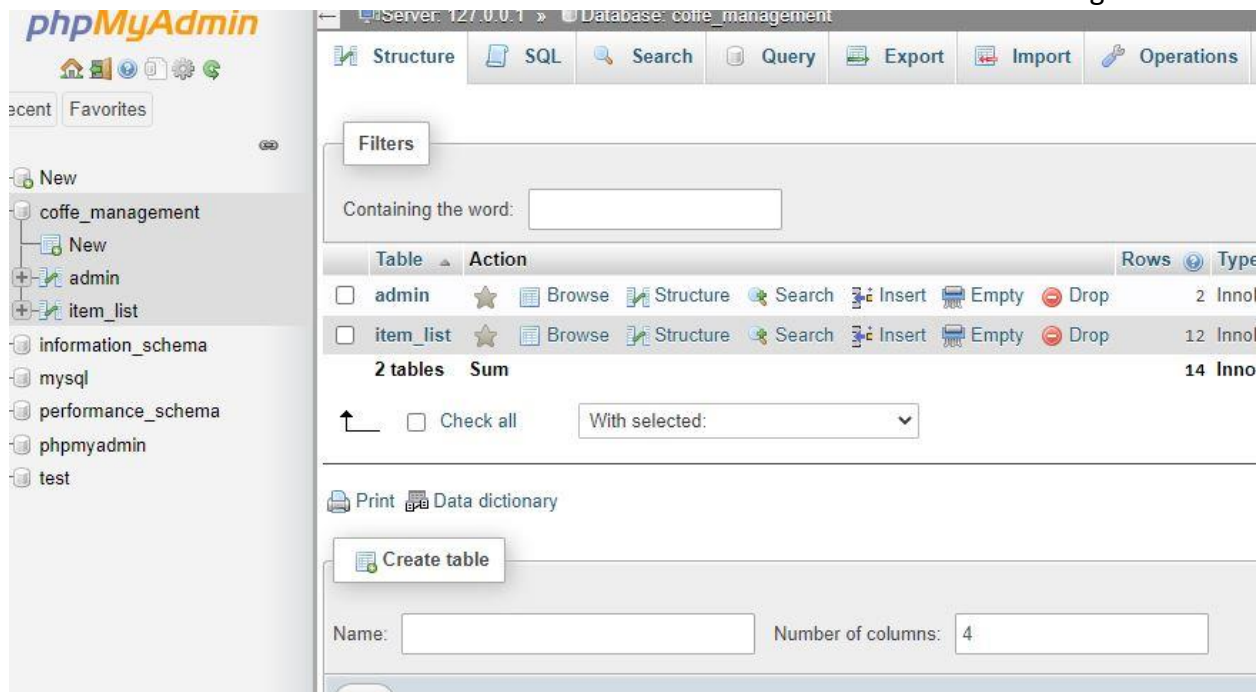
This chapter reader does know about our system and our interface. In previous chapter we told you about database, and our technology. We know that is too short. So let's start.

3.2 System Analysis

This section is especially for a user and admin. It's a manual of our system. So that, user can uses our system very easily.

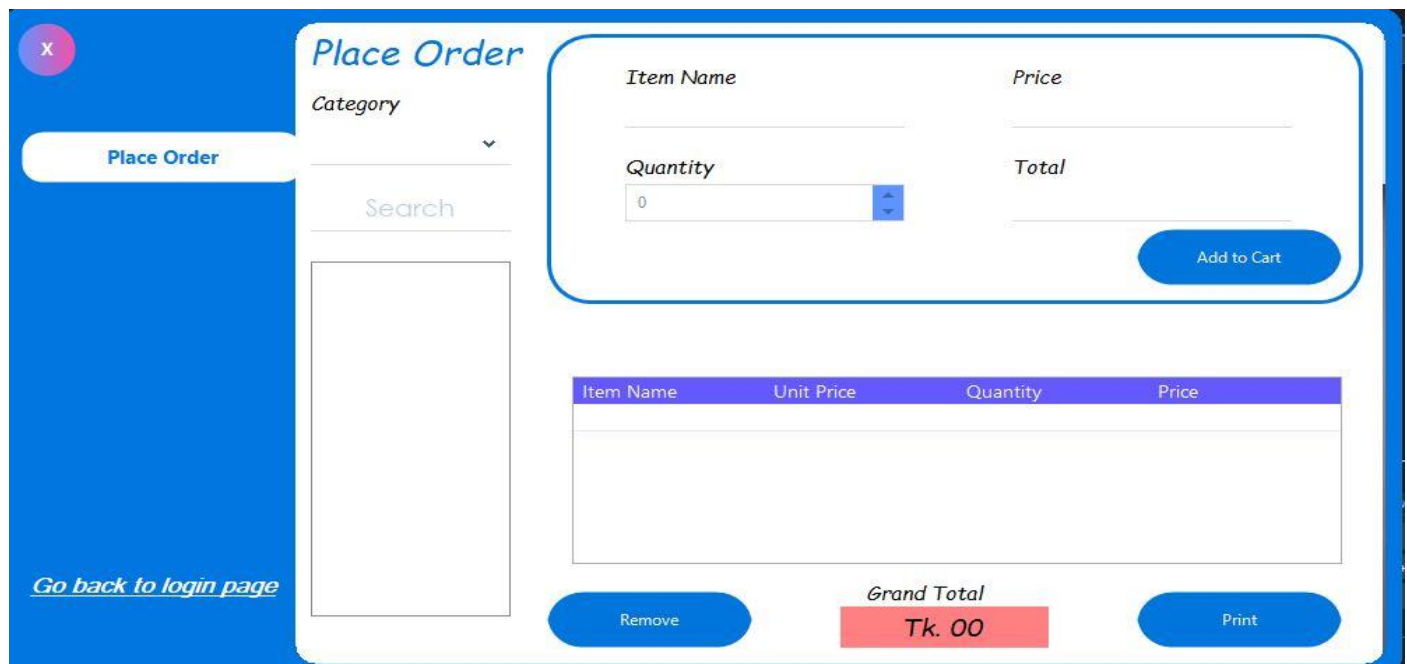
As an admin

An admin can access all the things. Like feature, products add or delete, employee's information and customer's information. We set a username and password for admin. So the admin doesn't create an account. Admin can also access the database. Products information, customer's information, employee's (waiter/chef) information, order and bill paid information will be saved in the database.



As a customer

A customer can only see the all item and ordered some food. Then they pay their bill and collect their receipt. Customer can search an items' category, price. Don't need any account for a customer to order food.



As an employee(waiter/chef)

An employee use same featurer of admin. Just an employee can't access the database. They don't know about customer information. But they can access of add product, update product, add order, employee information.

3.3 Interface

Here pictures of project:



Username

Password

Login

[Continue as Guest](#)



- Place Order
- Add Items
- Update Items
- Remove Items

[LogOut](#)



X

Place Order

Add Items

Update Items

Remove Items

LogOut

Place Order

Category

▼

Search

Item Name

Price

Quantity

Total

0

Add to Cart

Item Name	Unit Price	Quantity	Price
-----------	------------	----------	-------

Remove

Grand Total
Tk. 00

Print

X

Place Order

Add Items

Update Items

Remove Items

LogOut

Add New Item

category

▼

Item Name

Price

Add Item

X

Place Order

Add Items

Update Items

Remove Items

LogOut

Update Item

Item Name

Start Typing.....

21	Black Coffe	Coffee	45
22	Cold Coffee	Coffee	99
23	Coca Cola	Soft Drink	45
24	Sprite	Soft Drink	30
25	Maaza	Soft Drink	15
26	7Up	Soft Drink	60
27	Potato Chips	snacks	15
28	Popcorn	snacks	20

Category

Category

Price

Price

Item Name

Item Name

Update

X

Place Order

Add Items

Update Items

Remove Items

LogOut

Delete Item

Delete Item

Start Typing.....

21	Black Coffe	Coffee	45
22	Cold Coffee	Coffee	99
23	Coca Cola	Soft Drink	45
24	Sprite	Soft Drink	30
25	Maaza	Soft Drink	15
26	7Up	Soft Drink	60
27	Potato Chips	snacks	15
28	Popcorn	snacks	20
29	Beef Burgers	Burger	120
30	Chicken Buraers	Burraer	90

How to delete Items ?

3.4 Conclusion

In this chapter user can get knowledge about our project.

Future

4.1 Future Plan

4.2 Conclusion

4.1 Future Plan

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World moving very fast. Everyone depended on technology. Every big mill industry use technology. Technology used everywhere. Even education turning into technology based. So, we think to create this project. Our future plan is make this project user friendly and available on online.

4.2 Conclusion

So, this is our Cafe Management System project. It's very simple and user friendly.

✓ Chapter One is our introduction about this project. Users should be read this.

✓ Chapter Two is about database, XAMPP, Visual Studio 2019, .NET Framework,

MySQL. We think reader's can know about background of this project. Schema and ER diagram is our project map.

✓ Chapter Three is our system and interface of our project.

✓ Chapter Four is our future plan and conclusion of this project.

